

Freight rate prediction

Assessment | approach and model selection

Objective: predict posted freight rates in dollars per load. The company supplied 48,000 labeled January-October 2025 loads, 12,000 unlabeled November-December validation loads, and 31 fixed December scenarios. All original input rows are retained.

Chronological evaluation

January-June (28,806 rows) predicts July (4,912), January-July (33,718) predicts August (4,759). Model choices were fixed using these comparisons and the bounded error review. January-August (38,477) then predicts September-October (9,523). There is no holdout tuning or early stopping. Earlier descriptive exploration saw all development labels, the holdout was unused for selection, rather than analyst-unseen.

Why CatBoost, and which features?

July-August pooled MAE: mileage \$250.94, Ridge \$212.27, signal-free CatBoost \$122.39, December CatBoost \$127.37. CatBoost captures nonlinear physical and categorical effects. Both selected models use pickup, delivery, distance, equipment, weight and date, validation also uses four coordinates. Directed route, weekday/year cycles, elapsed days and an invalid-weight flag are derived. IDs, target, market_index and quote_signal are excluded. Signal definitions and prediction-time availability are unconfirmed. Learned preprocessing uses training rows only.

Fixed settings: 450 CatBoost trees, depth 6, learning rate 0.08, MAE loss, L2 regularization 5, seed 2025, CPU/two threads, remaining settings are explicit in the notebook. Negative weights become missing with an indicator. Earlier masking gains were only \$0.38/\$0.84 MAE, so the rationale is physical validity, not a proven large accuracy improvement. The historical Ridge benchmark retains negative weights for a like-for-like comparison.

Measured results and limitations

September-October holdout: 9,523 identical rows

Model	MAE (\$)	RMSE (\$)	R-squared
Signal-free CatBoost	120.04	638.51	0.82494
December CatBoost	136.65	642.78	0.82259
Mileage baseline	256.95	684.25	0.79896

September has 4,670 loads, October has 4,853. Signal-free monthly MAE is \$109.58 / \$130.11, December-feature MAE is \$120.33 / \$152.36, mileage MAE \$260.83 / \$253.22. Signal-free overall signed bias is -\$81.84. October's higher errors and broader underprediction are concerns, not reasons to recalibrate against the holdout.

Rare expensive loads remain difficult

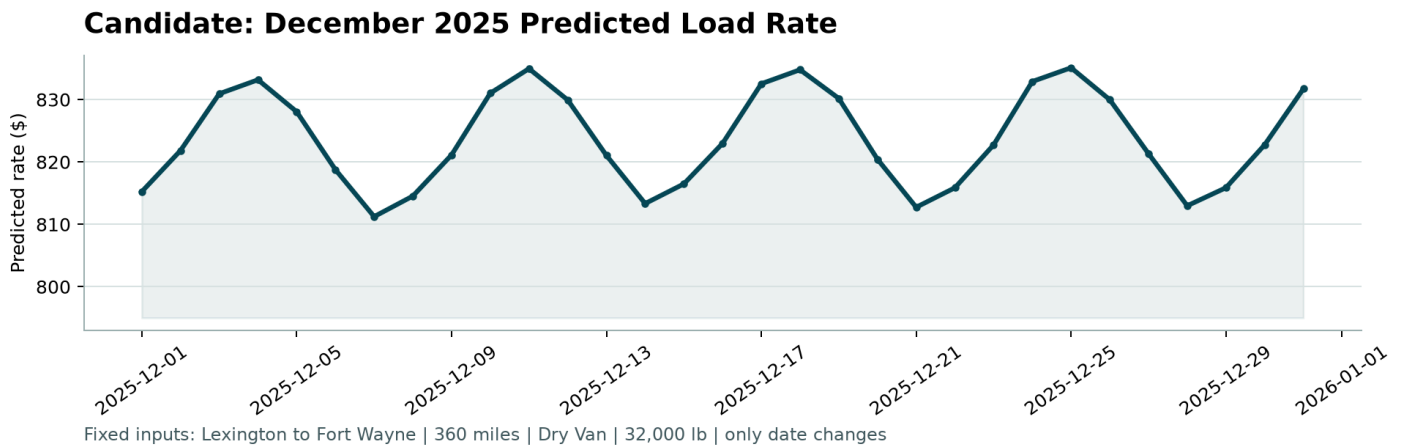
The training price P99 is \$5,968.61. All 101 holdout loads above that threshold are underpredicted. Their signal-free MAE is \$3,687, versus \$3,561 for mileage, they contribute 86.0% of squared error. The July-August review also found extreme rate-per-mile misses on known routes with ordinary weights. MAE loss favors typical prices. No implementation defect or confirmed bad label was found, no rows were dropped.

Practical limits

Only 21 holdout rows have unseen routes and 59 have negative weights, small subgroups cannot establish dependable generalization. Future validation has 12.175% development-unseen routes. One year of data and calendar trees offer limited extrapolation. Geographic provenance, special services, customer/carrier dependence and label delays are unresolved. Reduced-feature holdout accuracy does not validate December seasonality or the fixed route. Average improvement supports using the fixed models for this assessment, with these caveats.

December scenario and deliverables

Exact chart from the supplied, unchanged scorer



Lexington to Fort Wayne, 360 miles, Dry Van, 32,000 lb. Only date changes. December predictions range from \$811.23 to \$835.10. There are only 21 development loads on this route with this equipment, and no exact fixed-input match. The weekly pattern is a model estimate, neither December nor this route's accuracy has been directly validated.

Final training and execution

After holdout evaluation, both unchanged configurations were refitted on all 48,000 labeled rows. `validation_predictions.csv` contains 12,000 unique IDs in template order, `december_predictions.csv` contains all 31 dates with unchanged physical inputs. All predictions are finite and positive. The scorer accepts both files and creates `candidate_december.png`, it does not report hidden-label accuracy. Install `requirements.txt` and run `freight_rate_prediction.ipynb` from top to bottom. `README.md` and `loom_script.md` accompany this report. Nothing is submitted automatically.